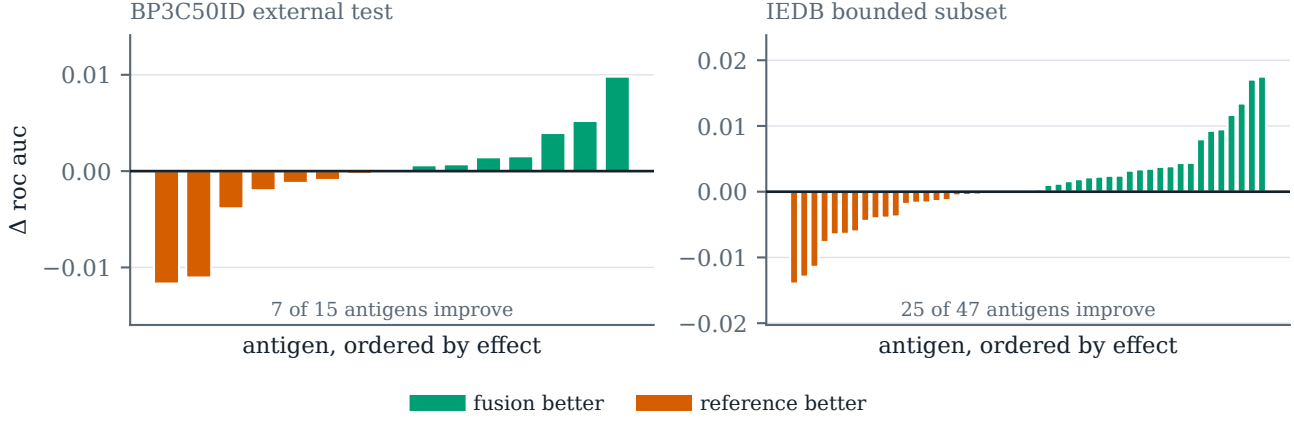Per-antigen fusion effect under leave-one-sequence-out selection